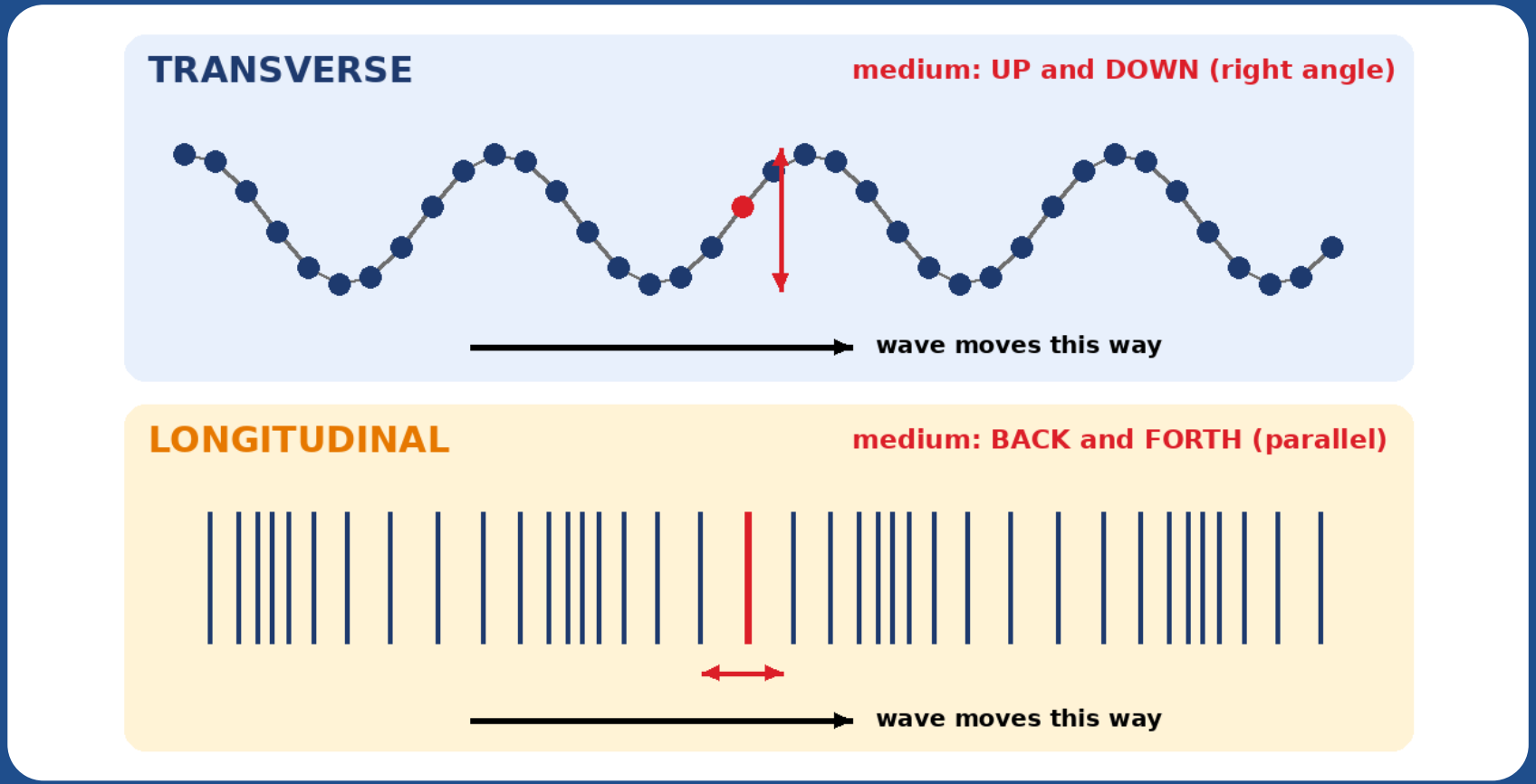
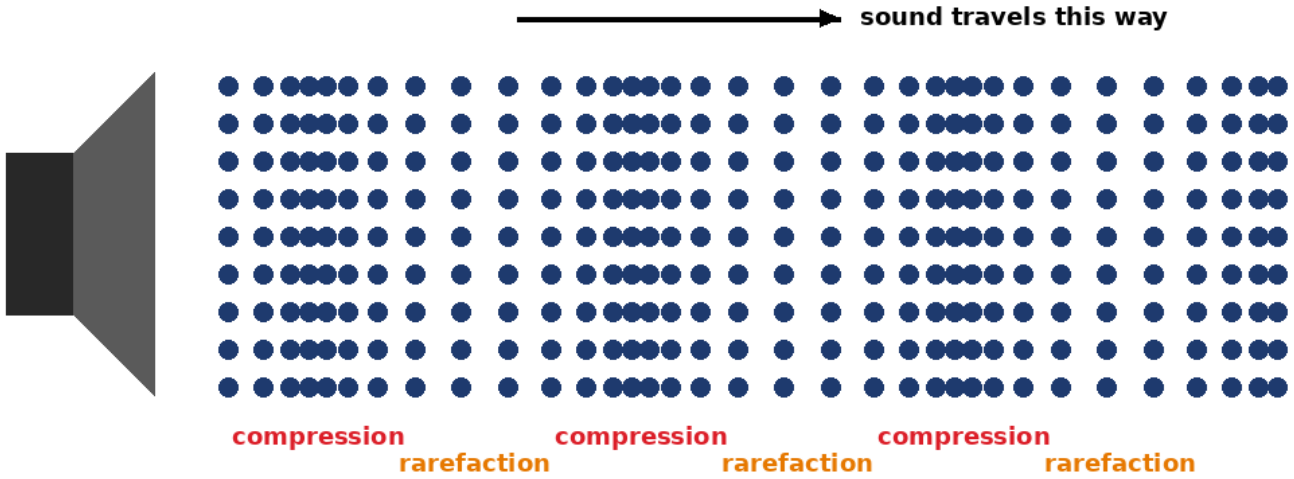
4.07 Longitudinal Waves

How does the medium move in a longitudinal wave?



Think About It



How does the sound of a drum get from the drum to your ear?

Single choice poll

A. Air from the drum flies to your ear

Single choice poll

B. The drum pushes air particles, and they bump the next ones

Single choice poll

C. Sound travels through empty space

Single choice poll

D. Light carries the sound

Today's Goals

I can describe how the medium moves in a longitudinal wave.

I can identify compressions and rarefactions.

I can explain why sound is a longitudinal wave.

I can compare transverse and longitudinal waves.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- Mechanical waves need a medium. The medium moves, then returns to its rest position.
- Vibrations make waves that carry energy. The energy of a mechanical wave increases with amplitude.
- Mechanical waves are transverse or longitudinal, depending on how the medium moves.

Inquiry skills (Grades 5–8): "Use appropriate mathematics, tools and techniques to gather data" and "Think critically and logically to connect evidence and explanations" (today's tuning fork, toy spring, and oPhysics activities).

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

In this lesson

- Integer warm-up
- Vocab check-in
- What is a longitudinal wave?
- Compare: transverse vs. longitudinal
- Sound is a longitudinal wave
- Toy spring: frequency and amplitude
- Sound wave diagram

CASE #4 • THURSDAY

Is the diver over 50 m above the whale?

WAVE WORD: *Mechanical wave*

A wave that needs a medium (such as water) to travel through.

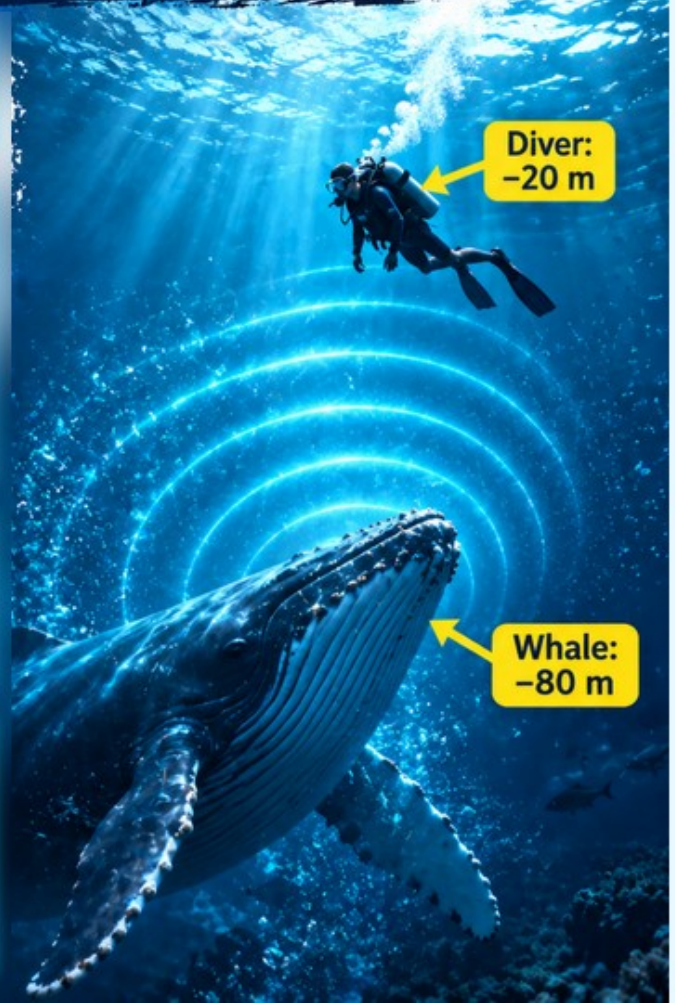
MYSTERY

Sound is a mechanical wave, so it needs a medium to travel through. A whale sings 80 meters below the surface, and the song travels through the water. A diver hears it from 20 meters below the surface. Is the diver more than 50 meters above the whale?

CLUES

- Whale: −80 m
- Diver: −20 m
- Diver − whale





CLAIM

Is the diver more than 50 m above the whale?

☐ YES ☐ NO

EVIDENCE

Solve:

−20 − (−80) = _____

Rewrite as adding: −20 + _____ = _____

REASONING

Finish the sentence:

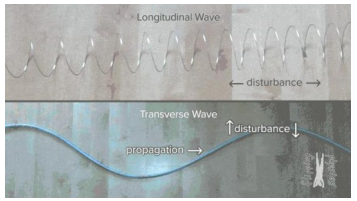
−20 − (−80) is the same as

−20 + _____ = _____.

The diver is _____ m above the whale.

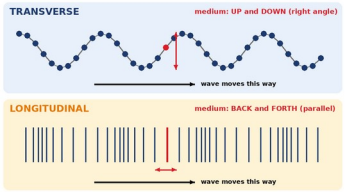
Vocab Check-In

From the 4.07 page. Starred words are on your Unit 4 vocabulary pretest.



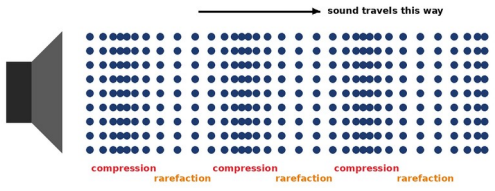
longitudinal wave

the medium moves back and forth, in the same direction as the wave



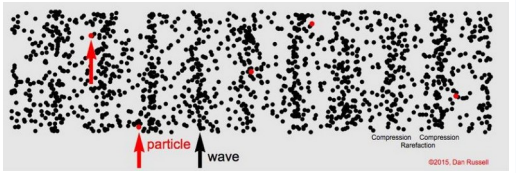
parallel

going the same direction, like two lanes on a road



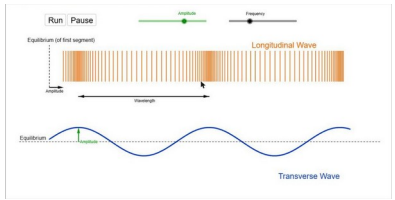
compression

where the particles are crowded close together



rarefaction

where the particles are spread farther apart



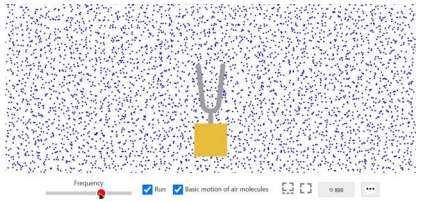
wavelength ★

in a longitudinal wave: compression to compression



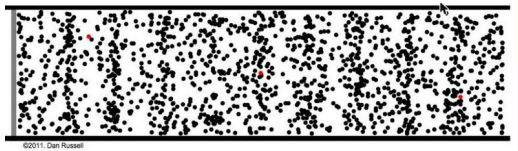
amplitude ★

how far particles move from rest; bigger = tighter compressions, more energy



sound wave

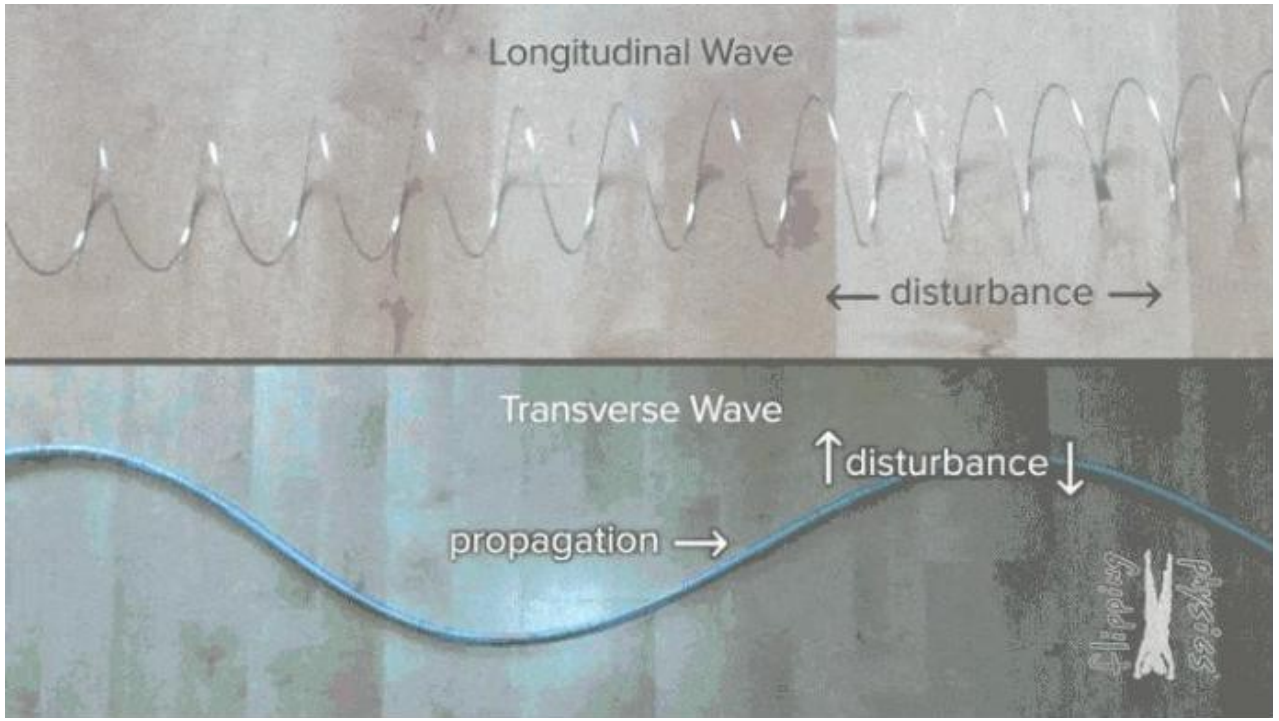
a longitudinal, mechanical wave made by vibrations



medium ★

the material a wave travels through: air, water, or a spring

What Is a Longitudinal Wave?

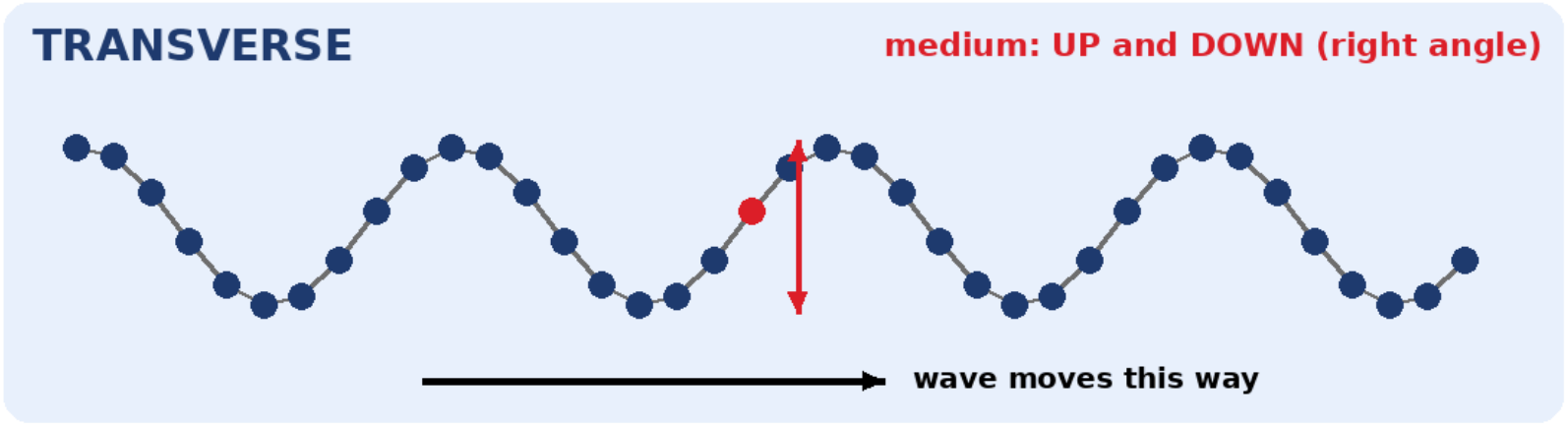


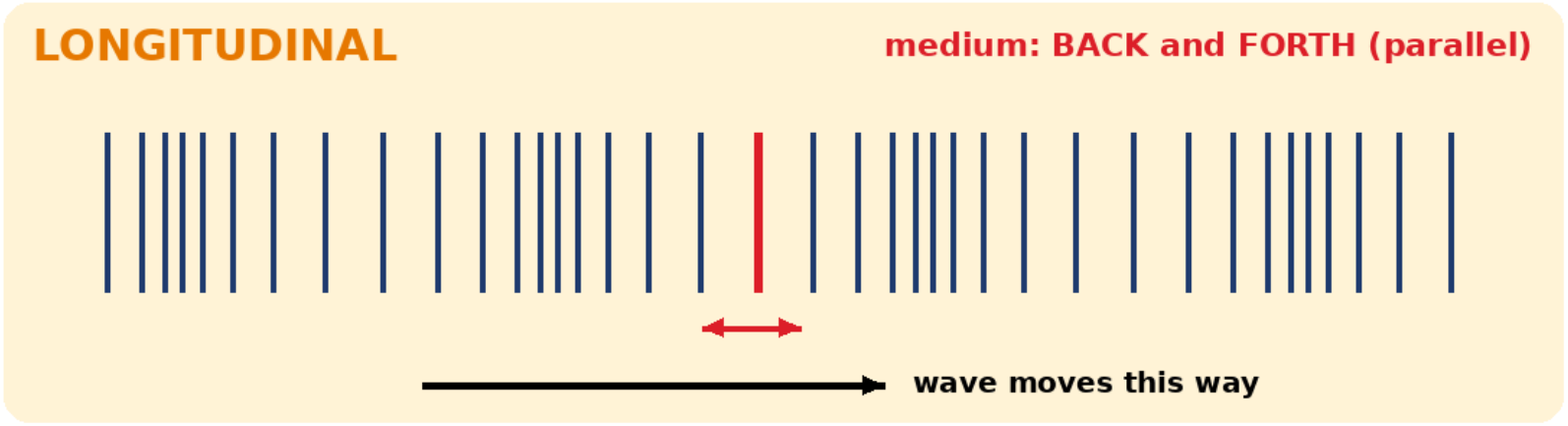
Longitudinal wave: the medium moves back and forth, in the SAME direction (parallel) as the wave.

Make one: stretch a slinky on the floor, then push and pull one end.

No hills or valleys. You see bunched-up and spread-out coils instead.

Compare: Watch the Red Coil





Different

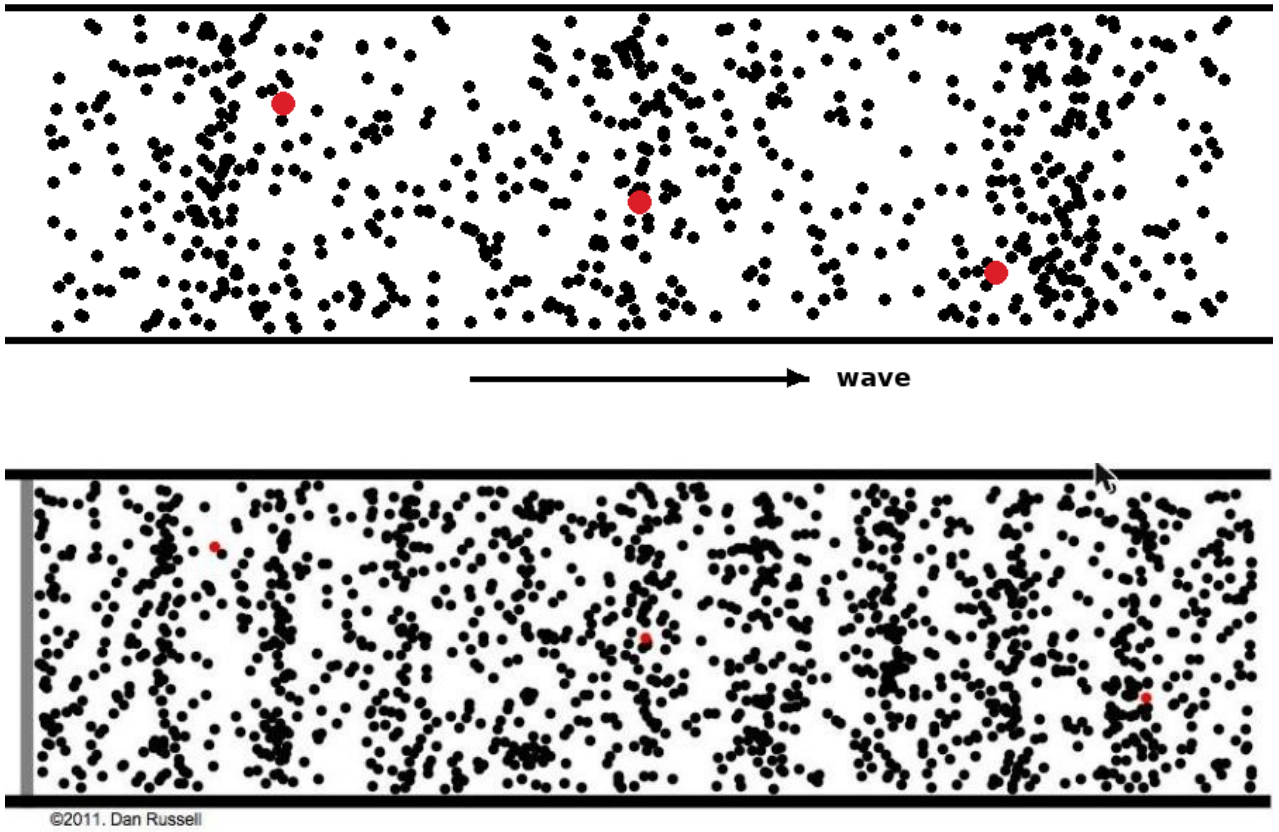
- Transverse: up and down, at a right angle.
- Longitudinal: back and forth, parallel.

Same

- The red coil ends up where it started.
- Only the energy travels.

Turn and talk: What is the SAME about both waves? What is DIFFERENT?

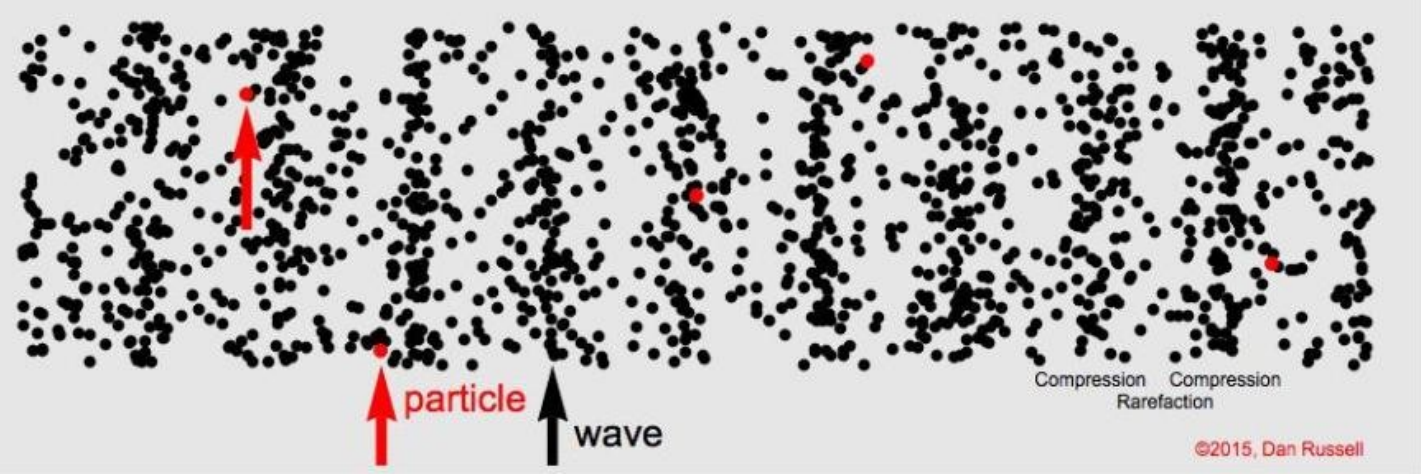
Prove It! Follow One Red Dot

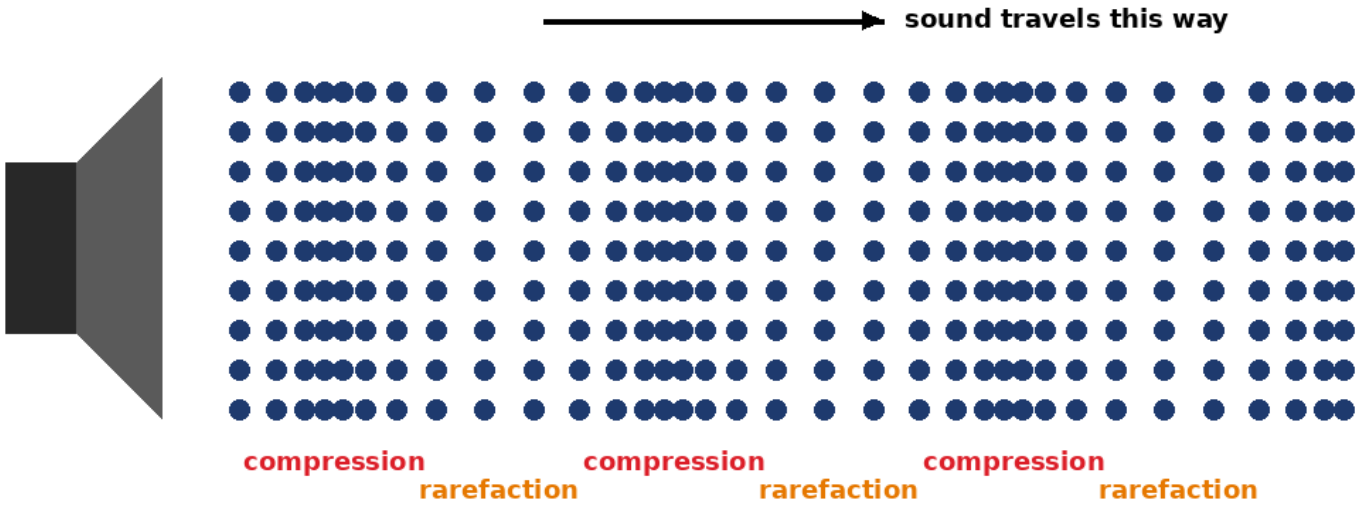


Does the red dot travel across the box, or just move back and forth?

The red dots move back and forth and stay near where they started. The bunched-up areas move across the box. That is the energy moving, not the matter.

Compressions and Rarefactions





Compression: particles crowded together.

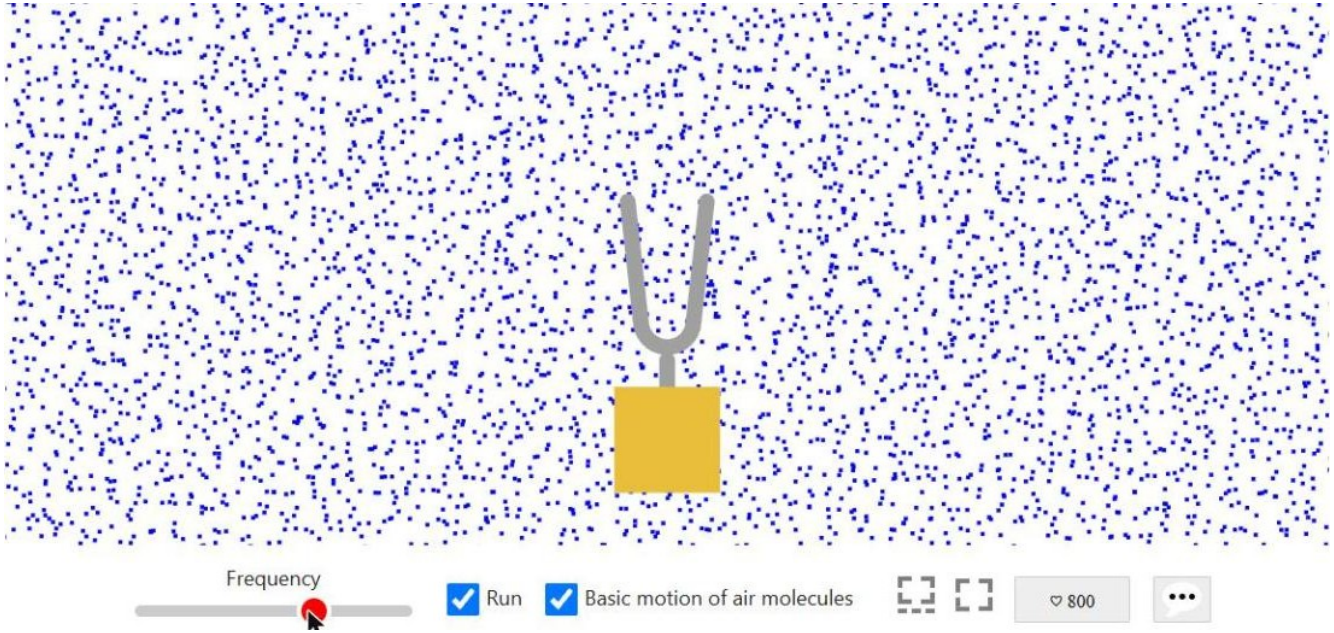
Rarefaction: particles spread apart.

The black arrow (wave) rides along. The red arrow (particle) just moves back and forth.

▶️ **Watch: Longitudinal Waves (1 min)**

The narrator says "amplification." The science word is amplitude.

Part 1: Sound Is a Longitudinal Wave



▶️ **Open Tuning Fork and Sound Wave**

Check Basic motion of air molecules. Slide Frequency up and down. JavaLab shows ads.

A drum or tuning fork vibrates. It pushes and pulls the air next to it. Those air particles bump the next ones, so compressions and rarefactions travel to your ear.

Sound is mechanical, so it needs a medium. There is no sound in empty space.

Engageli Poll

An astronaut bangs a hammer outside the space station. Why can't anyone hear it?

Single choice poll

A. Space is too cold

Single choice poll

B. Sound needs a medium, and space has almost none

Single choice poll

C. Sound is a transverse wave

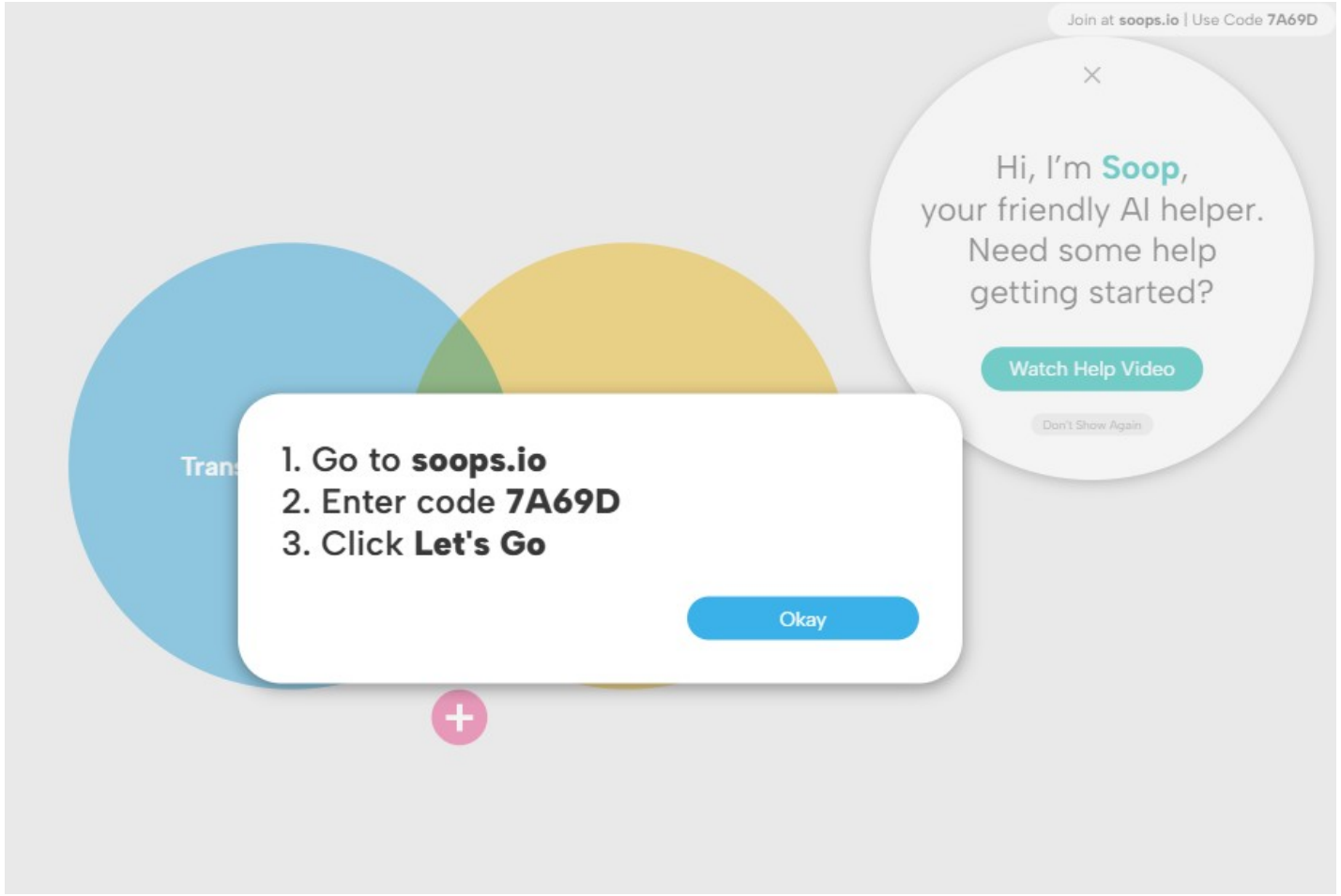
Single choice poll

D. Sound moves too fast in space

Compare: How They Move

	Transverse wave	Longitudinal wave
How the medium moves	Up and down	Back and forth
Compared to the wave	At a right angle (perpendicular)	Same direction (parallel)
Shape	Hills and valleys	Bunched-up and spread-out areas
Does matter travel?	No. Only energy travels.	No. Only energy travels.
Needs a medium?	Mechanical ones do. Light does not.	Yes, always.
Examples	shaken rope, guitar string, light	slinky pushed in and out, sound

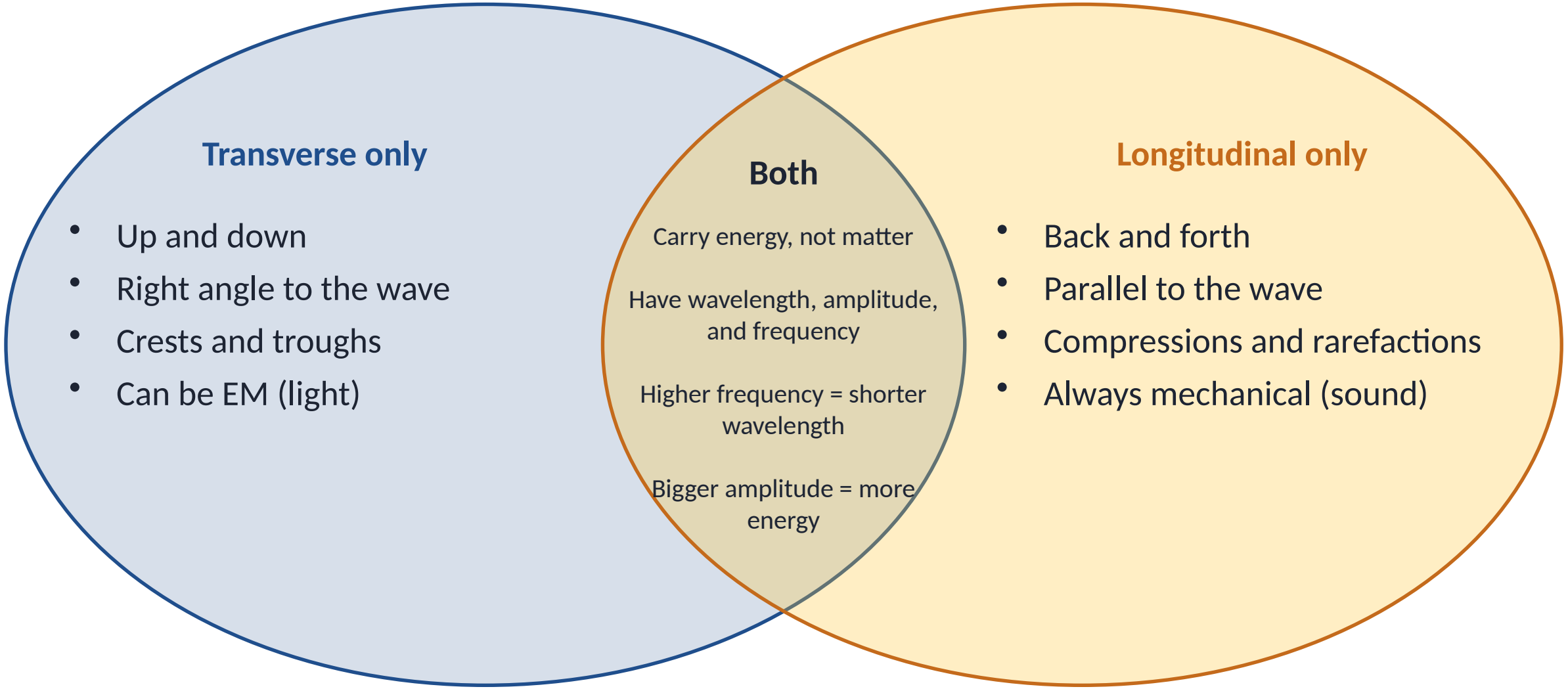
Your Turn: Build the Venn Diagram



1. Go to **soops.io**
2. Enter code **7A69D**
3. Click **Let's Go**

Add one fact to Transverse, Longitudinal, or the middle (both).

Compare: Venn Diagram



Engageli Poll

Which one is a LONGITUDINAL wave?

Single choice poll

A. a plucked guitar string

Single choice poll

B. sunlight

Single choice poll

C. a slinky pushed in and out

Single choice poll

D. a rope shaken up and down

Part 2: Longitudinal Waves in a Toy Spring



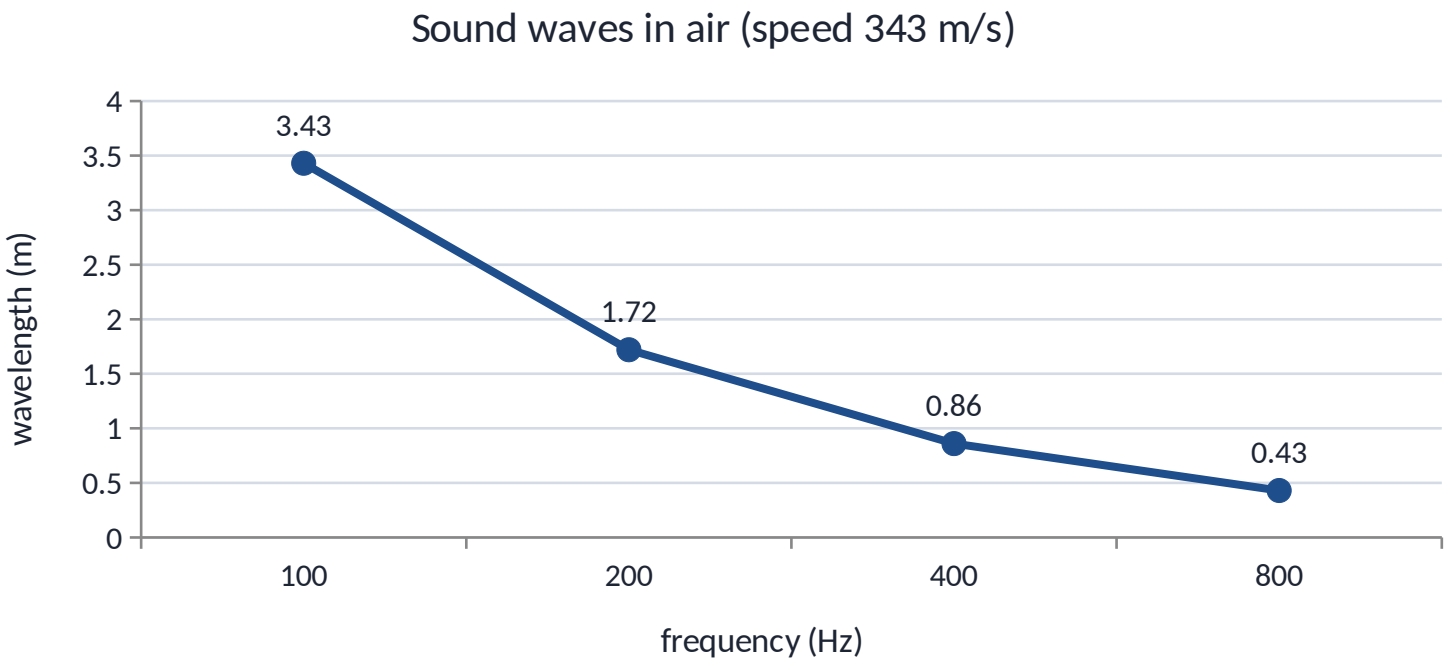
Frequency	Wavelength
2 Hz	___ cm
4 Hz	___ cm
5 Hz	___ cm

Watch ONE coil. Then move only the Frequency slider and read the Wavelength under the Pause button.

▶️ **Open the toy spring sim**

Same toy spring as 4.06, but now the coils move back and forth.

Graph Poll



Read the graph.

What happens to the wavelength of sound as the frequency goes up?

Single choice poll

A. It gets longer

Single choice poll

B. It gets shorter

Single choice poll

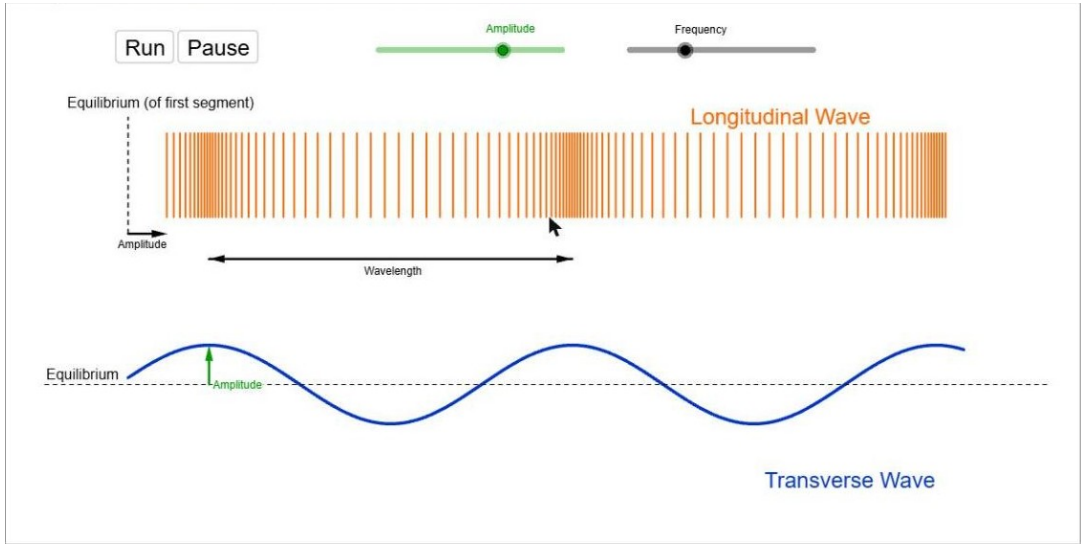
C. It stays the same

Single choice poll

D. It goes up, then down

Classkick: 4.07 Parts 1–2 • Q4

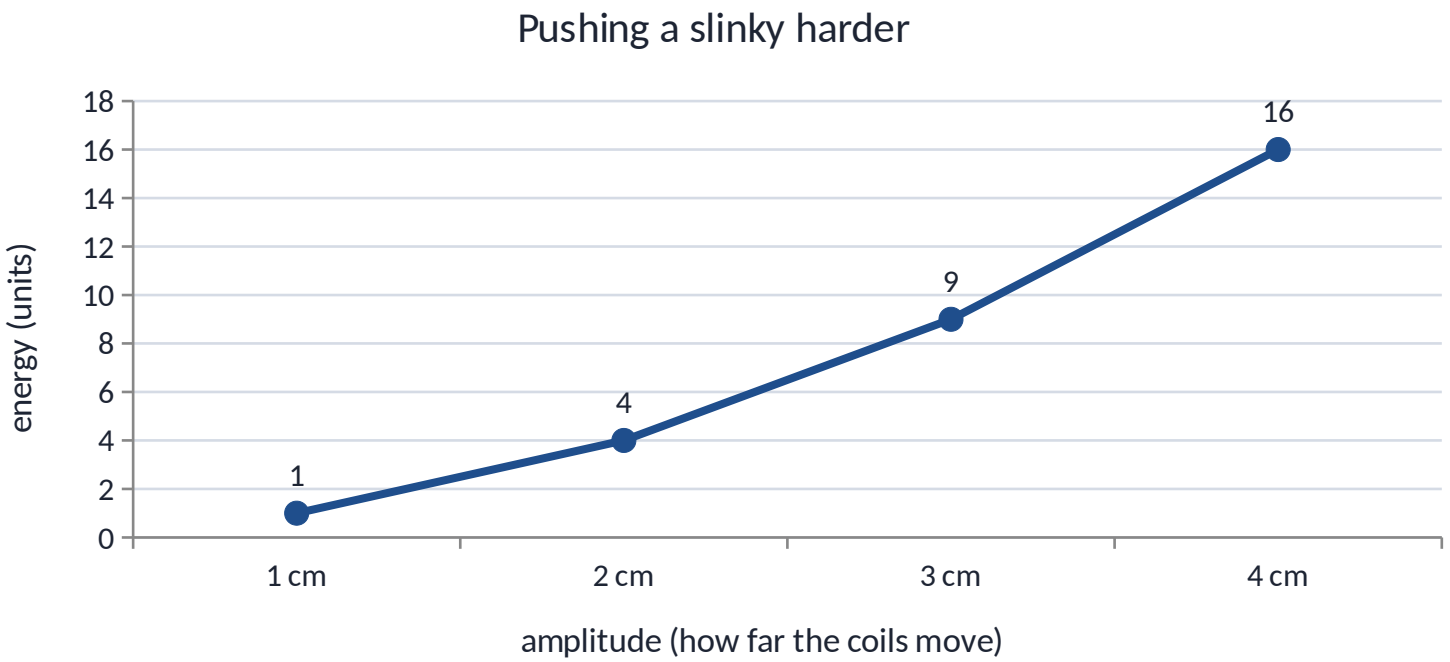
Compare: Change the Amplitude and Frequency



	Transverse	Longitudinal
Bigger amplitude	Taller crests, deeper troughs. More energy.	Tighter compressions, wider rarefactions. More energy.
Higher frequency	Crests closer together: shorter wavelength.	Compressions closer together: shorter wavelength.
Wave speed (same medium)	Stays the same	Stays the same

▶️ Open oPhysics: both waves

Graph Poll



Read the graph.

What happens to the energy as the amplitude gets bigger?

Single choice poll

A. It goes down

Single choice poll

B. It stays the same

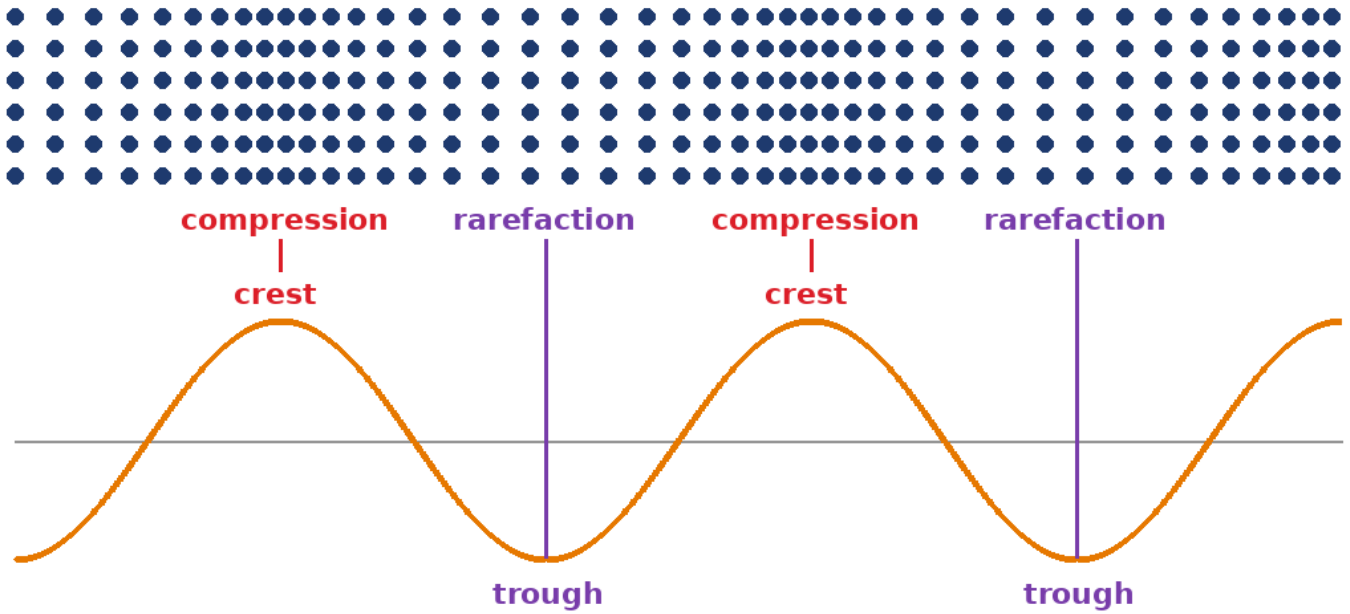
Single choice poll

C. It goes up

Single choice poll

D. It goes up, then down

Compare: Drawing Sound as a Graph

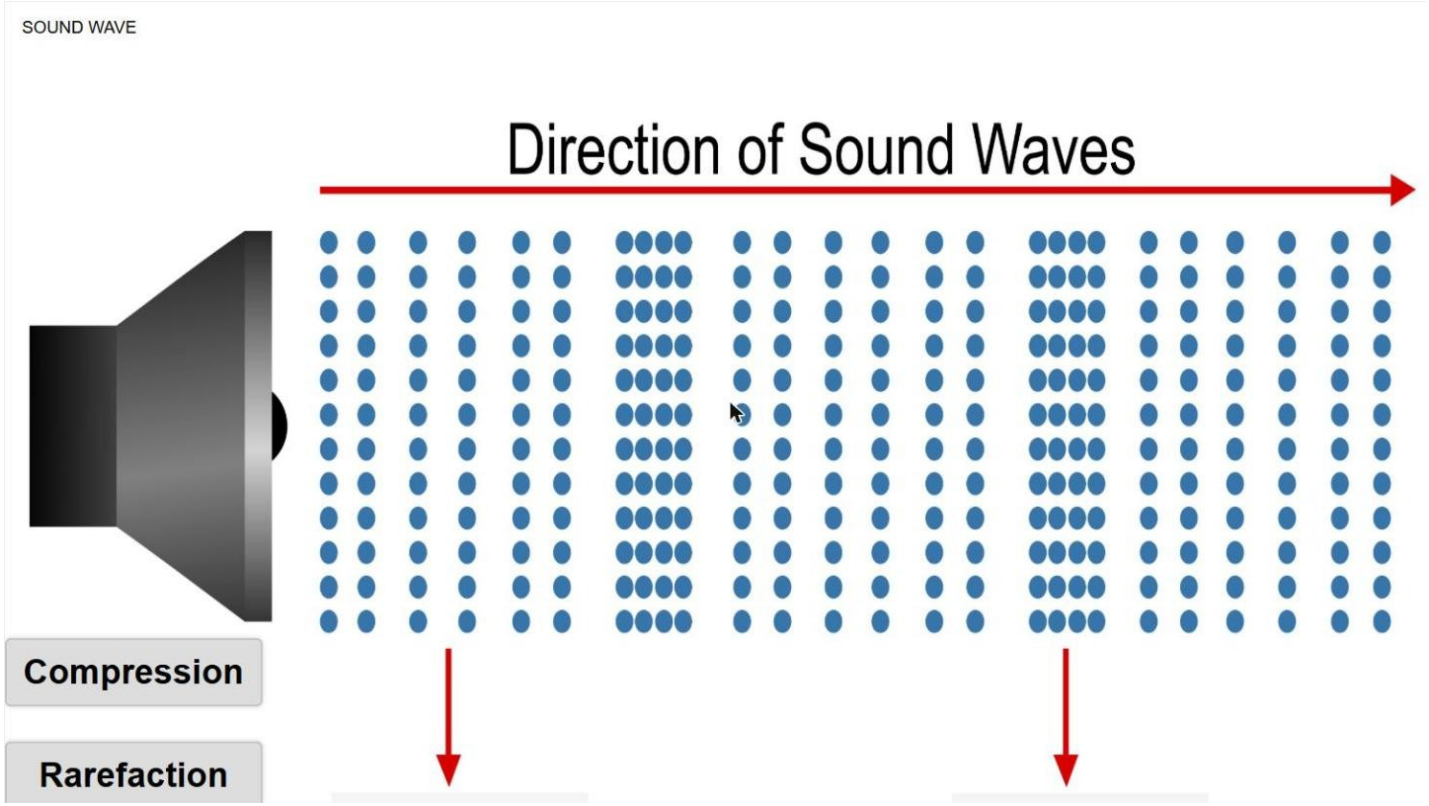


Top: the air particles in a sound wave. Bottom: a graph of the same wave.

	Transverse	Longitudinal
Parts	crests and troughs	compressions and rarefactions
One wavelength	crest to crest	compression to compression
On a graph	crest = high point	compression = crest, rarefaction = trough

Scientists often draw sound like a transverse wave. The graph is a model. The air still moves back and forth.

Part 3: Practice Wave Parts



▶️ Open the Sound Wave Diagram

Drag each label onto the diagram and press Check.

Bunched-up dots = **compression** (a crest on the graph).
Spread-out dots = **rarefaction** (a trough on the graph).

One wavelength: compression to the next compression.

Who Is Right?

Mia draws sound as bunched-up lines. Leo draws it as a wavy line with crests and troughs. Who drew sound correctly?

Ava says:

"Only Mia. Sound is longitudinal, so a wavy line is wrong."

Ben says:

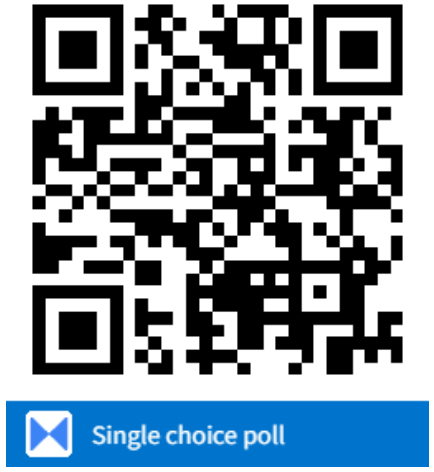
"Only Leo. Every wave has crests and troughs."

Cam says:

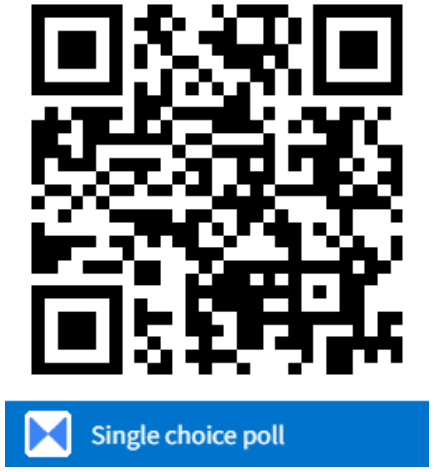
"Both. The wavy line is a graph: crests show compressions and troughs show rarefactions."

Dee says:

"Neither. Sound is invisible, so it cannot be drawn."



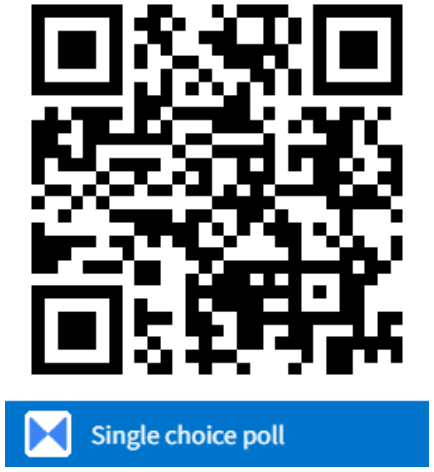
A. Ava



B. Ben



C. Cam



D. Dee

Engageli Poll

Can light ever be a longitudinal wave?

Single choice poll

A. Yes, light is always longitudinal

Single choice poll

B. No. Light is EM, and EM waves are always transverse

Single choice poll

C. Only when it travels through water

Single choice poll

D. Only in empty space

Key Ideas

1. Longitudinal wave: the medium moves back and forth, parallel to the wave.
2. Compressions are crowded areas. Rarefactions are spread-out areas.
3. Sound is a longitudinal, mechanical wave. It needs a medium.
4. Both kinds carry energy, not matter. On a graph, compression = crest and rarefaction = trough.

Re-rate your goals. Then start the 4.07 Quest Map (#32).